



**THE UNIVERSITY OF AZAD JAMMU AND
KASHMIR, MUZAFFARABAD**

Department of Software Engineering



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LAB:

01

COURSE TITLE:

Database Systems

COURSE CODE:

CS-2204

INSTRUCTOR:

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LAB 01: INSTALLATION OF XAMPP & INTRODUCTION TO PHPMYADMIN

1. LAB OBJECTIVES

1. Download and install XAMPP on a local Windows machine.
2. Verify the successful startup of Apache and MySQL services via the XAMPP Control Panel.
3. Access and navigate the phpMyAdmin web interface.
4. Create a new database named Blog_website through phpMyAdmin.
5. Create a user's table as a practical introduction to database schema design.

2. TOOLS & ENVIRONMENT

Tool / Component	Description
XAMPP (latest stable)	Local server stack bundling Apache, MySQL, PHP, and phpMyAdmin in a single installer
Apache HTTP Server	Web server included in XAMPP; serves pages locally on port 80 / 443
MySQL 8.x	Relational database engine bundled with XAMPP; runs on port 3306
phpMyAdmin	Browser-based GUI for managing MySQL databases without writing raw SQL
Windows 11	Host operating system on which XAMPP was installed
Google Chrome	Browser used to access the phpMyAdmin interface at localhost

3. BACKGROUND & THEORY

XAMPP is a free, open-source local server stack by Apache Friends. The acronym stands for Cross-platform, Apache, MariaDB/MySQL, PHP, and Perl. It provides all components required to run a local database server in a single installer — the standard setup for academic database labs.

phpMyAdmin is a browser-based GUI for managing MySQL databases without writing raw SQL. In any database-backed application, the database stores all persistent data. Configuring the local environment and creating the foundational schema is always the first step — which is what this lab demonstrates.

4. STEP-BY-STEP PROCEDURE

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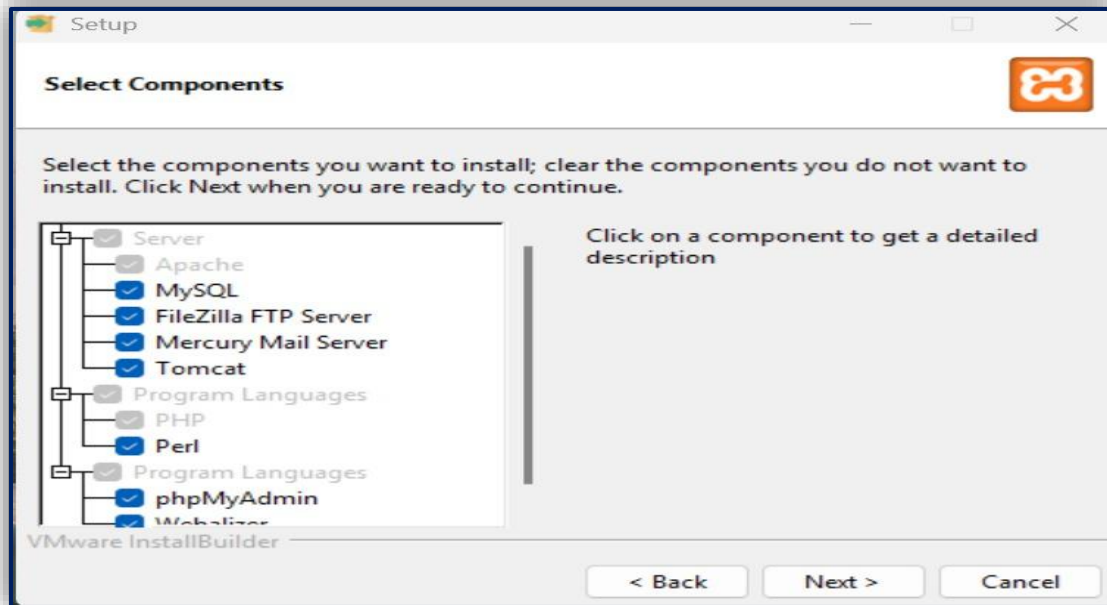
STEP 1 — DOWNLOADING XAMPP

- Visited <https://www.apachefriends.org> and downloaded the Windows .exe installer.



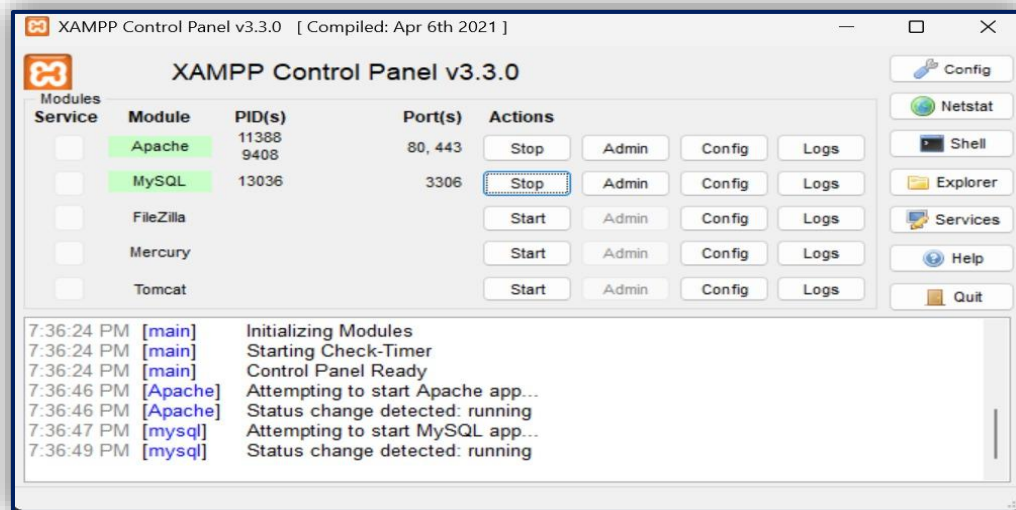
STEP 2 — INSTALLING XAMPP

- Ran installer as administrator; selected Apache, MySQL, PHP, phpMyAdmin components.
- Used default directory C:\xampp and completed the setup wizard.



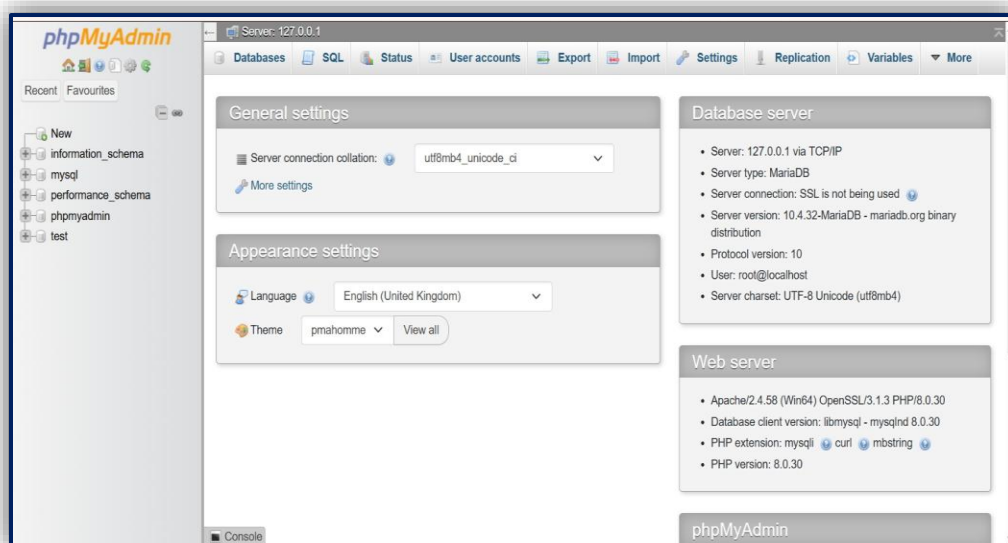
STEP 3 — STARTING APACHE & MYSQL

- Opened XAMPP Control Panel; clicked Start for Apache (port 80) and MySQL (port 3306).
- Both modules showed green status confirming successful startup.



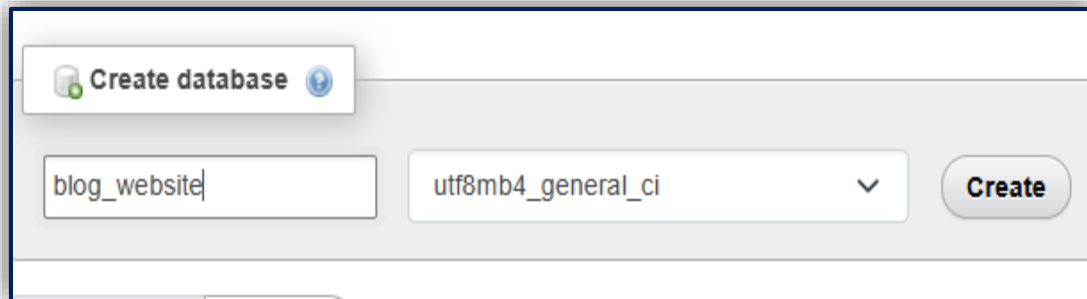
STEP 4 — ACCESSING PHPMYADMIN

- Navigated to <http://localhost/phpmyadmin> in Chrome; dashboard loaded successfully.



STEP 5 — CREATING THE BLOG_WEBSITE DATABASE

- Clicked New → entered name Blog_website → collation utf8mb4_general_ci → clicked Create.



STEP 6 — CREATING THE USERS TABLE

- Selected Blog_website → entered table name users with 5 columns → clicked Save.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id 🔑	int(11)			No	None		AUTO_INCREMENT
2	first_name	varchar(100)	utf8mb4_general_ci		No	None		
3	last_name	varchar(100)	utf8mb4_general_ci		No	None		
4	email 📧	varchar(255)	utf8mb4_general_ci		No	None		
5	password	varchar(255)	utf8mb4_general_ci		No	None		
6	role_id 🔑	int(11)			No	3		
7	profile_img_url	varchar(255)	utf8mb4_general_ci		Yes	NULL		
8	created_at	timestamp			No	current_timestamp()		

5. OBSERVATIONS & RESULTS

XAMPP INSTALLATION

Installed without errors; Apache and MySQL both started successfully with green indicators.

PHPMYADMIN ACCESS

Interface loaded at localhost/phpmyadmin immediately after services started; no extra authentication required.

DATABASE CREATION

Blog_website created with utf8mb4 collation, confirming full Unicode support for a real-world blog.

TABLE CREATION

Users table created with surrogate key, unique email constraint, and auto timestamp — a standard user-auth schema pattern.

6. CONCLUSION

This lab successfully established a fully functional local database development environment using XAMPP. Apache and MySQL services were configured and verified, and phpMyAdmin was confirmed accessible via the browser. A practical exercise was performed by creating the Blog_website database and a users table, demonstrating the complete workflow from environment setup to schema creation. This foundation will support all subsequent labs in this course.